

# Correct Answers, Uneven Grounding

The distributed-evidence gap in legal RAG

Answers remain similarly accurate. Complete source support falls when evidence spans pages.

-12.2 pp  
page-grounding gap  
95% CI -19.0 to -5.9

95 eligible questions within a 100-question benchmark / 30 PDFs / 590 pages

## 1 HYPOTHESIS

A model can answer from retrieved evidence yet emit an incomplete source set. Aggregate scores can hide whether this mismatch depends on evidence shape. [1,4,5]

**RESEARCH QUESTION**  
How does distributed evidence affect the relationship between answer quality and source grounding in a legal RAG pipeline?

**H1**  
Multi-page evidence will preserve answer quality but reduce final page grounding. Uniform citation narrowing may widen that gap.

59 localized exactly one gold page 36 distributed multiple gold pages

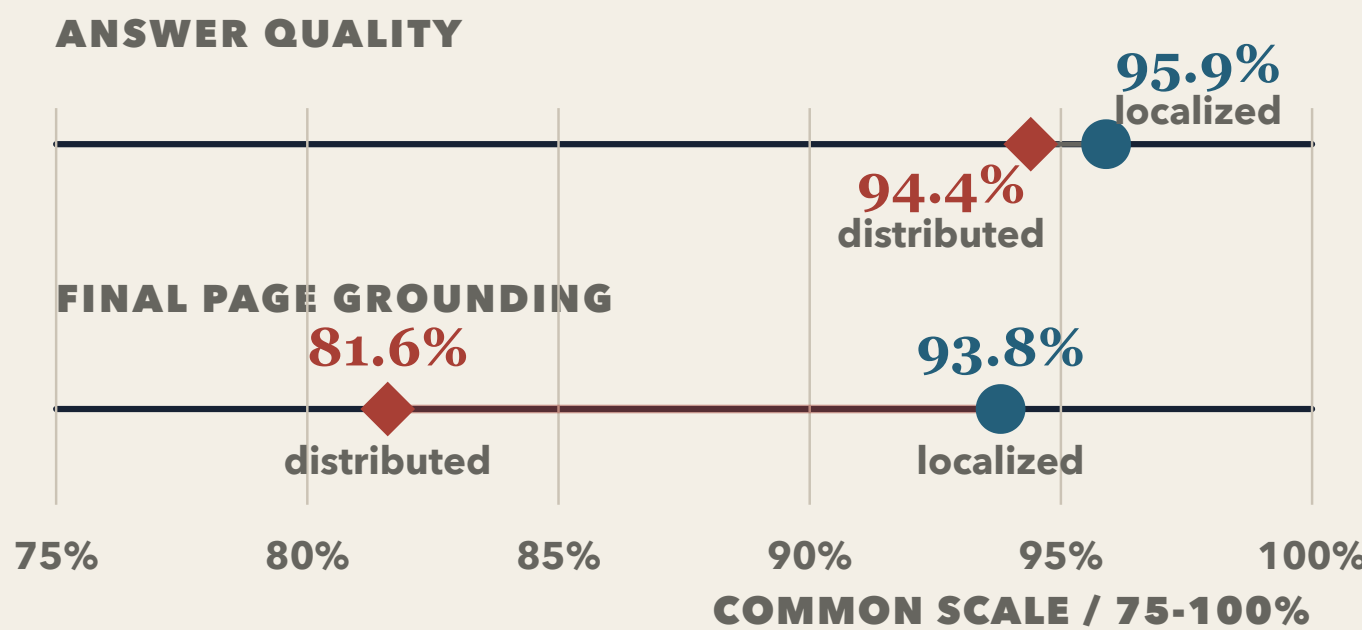
**Answer quality** / mean benchmark score on a 0-1 scale.  
**Source grounding** / page F-beta, beta = 2.5, weighting recall.  
**Evidence shape** / 35 of 36 distributed questions also span documents; the effects cannot be separated.

## 2 METHODOLOGY

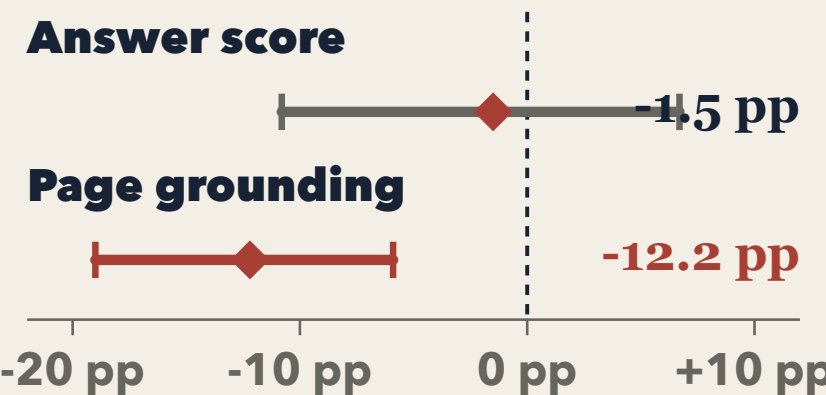


## 3 RESULTS

Answers stay level.  
Grounding separates.



### DISTRIBUTED MINUS LOCALIZED



20,000 stratified bootstrap resamples. The answer interval crosses zero; the grounding interval does not.

### PERFECT ANSWER / GROUNDING BELOW 0.8

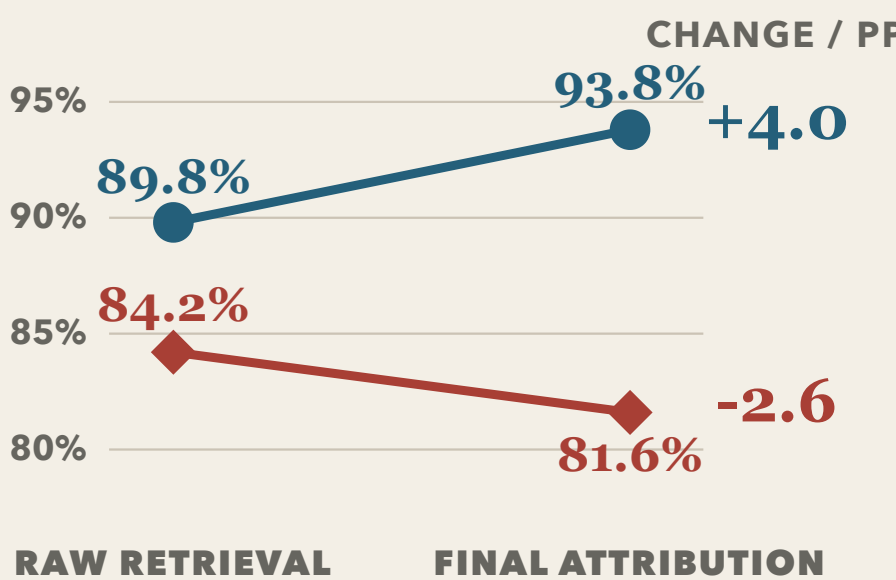
Localized 4 / 55 7.3%  
Distributed 16 / 34 47.1%

Among answers scoring 1.0, incomplete page support is much more common for distributed evidence.

## 4 RESULTS / DIAGNOSTIC

Citation narrowing moves groups apart.

Page-level F-beta from the same run, before and after answer-conditioned attribution.



### DECISION

Use evidence spread as a routing signal: narrow localized support; preserve and aggregate distributed support.

Gold evidence shape is unavailable at inference time. Use observable proxies: retrieval spread, cross-document dependencies, or claim-level support checks.

## LIMITATIONS

One saved development run; retrospective hypothesis; observational groups; page and document spread are confounded; answer types and scoring methods are uneven; the grounding-below-0.8 threshold is descriptive; stage changes are not a causal ablation.

## REPRODUCIBILITY

[github.com/SteinVR/Agentic-RAG-Challenge](https://github.com/SteinVR/Agentic-RAG-Challenge)  
20,000 stratified bootstrap resamples / seed 20260903. Saved-run artifacts include `solver_narrowing_true.yaml`, `benchmark_answer_scores.checkpoint.jsonl`, and `grounding/ledger.csv`.

## SELECTED REFERENCES

[1] Lewis et al., NeurIPS 2020 / RAG. [2] Karpukhin et al., EMNLP 2020 / dense retrieval. [3] Cormack et al., SIGIR 2009 / RRF. [4] Gao et al., EMNLP 2023 / citations. [5] Pipitone and Houir Alami, 2024 / LegalBench-RAG. [6] Zhang et al., 2025 / Qwen3 Embedding.